Commutative Algebra

Structure of the course

1. Ring theory: ideals
2. Module theory
3. Localization, integral dependence & going-up, going-down theorems
4. Chain conditions; Noetherian rings.

   primary decompositions.

5. Artinian rings & Hilbert's null-stellensatz

Reference: Introduction to commutative Alg by MF Atiyah & IG Macdonald.

Ring Recall

Def^n: A ring A is a set with two binary operations, called addition $(+)$ & multiplication s.t.

① $(A,+)$ is an abelian group.

② the multiplication is associative & is distributive over addition.

③ For $x,y \in A$, $xy = yx$.

④ $\exists\, 1_A \in A$ s.t. $1_A x = x \quad \forall x \in A$

Examples: ① $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$

(0) : $A = (0)$ is a commutative ring with $1_A = 0$.

A is called the zero ring.

② If A is a commutative ring, then so is $A[x] = \{a_0 + a_1 x + \cdots + a_n x^n : n \in \mathbb{N}, a_i \in A\}$

$A[x_1, \dots, x_k] = \{\sum a_{i_1} \cdots a_{i_k} x_1^{i_1} \cdots x_k^{i_k} : i_j \in \mathbb{N} \cup \{0\}\}$

③ S be any set

$$\mathcal{F}(S) := \{ f : S \to \mathbb{R} \}$$

$$(f+g)(x) := f(x) + g(x)$$

$$(fg)(x) := f(x) \cdot g(x)$$

<u>Def</u>: <u>Subring</u>

Let A be a ring. A subset $B \subseteq A$ is a subring, if B itself is a ring w.r.t. same binary operations on A.

<u>Examples</u>: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

$*$ $A \subseteq A[x]$

$$A \hookrightarrow A_1 \subseteq A[x]$$

$$A_1 = \{ p(x) : p(0) = 0 \} = \langle x \rangle$$

Homomorphism:

A map $f: A \longrightarrow B$ is said to be a ring homomorphism if

(1) $\forall x, y \in A$, $f(x+y) = f(x) + f(y)$

(2) " $f(xy) = f(x) f(y)$

If $1_A \in A$, then $f(1_A) = 1_{f(A)}$

Observation: (1) $\operatorname{Im} f$ is a subring of B

(2) $\ker f = \{a \in A : f(a) = 0\}$ is a subring of A.

Ideals: A subset $I \subseteq A$ is an ideal of A if $(I, +)$ is an abelian group & for every $a \in A$ & $x \in I$, $ax \in I$.

Example:

(1) $\{0\} \subseteq A$

(2) $n\mathbb{Z} \subseteq \mathbb{Z}$

(3) $\langle 0 \rangle \subseteq \mathbb{Q}$

④ $\langle x \rangle \subseteq A[x]$

⑤ $A \subseteq F(S)$

$$A = \{ f \in F(S) : f(x) = 0 \ \forall x \in X \neq \varphi \}, \quad X \subseteq S$$

Ex: If $f : A \to B$ is a ring homomorphism then $\ker f$ is an ideal of A.

$a \sim b \iff a - b \in I$ is an equivalence relation. & the collection of all equivalence classes denoted by A/I, is called quotient of A wrt I.

Define: $(a+I) + (b+I) = (a+b) + I$

$(a+I)(b+I) = ab\, I$.

Then A/I is a commutative ring with identity $(1_A + I)$ w.r.t. the above binary operation.

Observe: $\exists$ a natural map

$$\varphi : A \longrightarrow A/I \text{ given by } \varphi(a) = a+I.$$

& φ is surjective ring homomorphism.

Isomorphism thm:

If $f : A \to B$ is a ring homomorphism

then, $\boxed{\operatorname{Im} f \cong A/\ker f}$

Ideals of A/I are in one-one correspondence with ideals of A containg I.

if $J \trianglelefteq A \Rightarrow \varphi(J) \overset{?}{\trianglelefteq} A/I \quad [\varphi : A \to A/I]$

$x \in A/I \Rightarrow \exists\, a \in A$ s.t. $\varphi(a) = x$.

$x\varphi(J) = \varphi(a)\varphi(J)$

$= \varphi(aJ) = \varphi(J)$

$\varphi(J) = \{ j+I \mid j \in J \} = J/I \quad I \trianglelefteq J$

$\mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/3\mathbb{Z}$

$a \longmapsto a + 3\mathbb{Z}$

$$\varphi(2\mathbb{Z}) = \{2m + 3\mathbb{Z} \mid m \in \mathbb{Z}\}$$
$$= \mathbb{Z}$$

$f : A \longrightarrow B$ ring homomorphism.

$I \subseteq A$ an ideal $\Rightarrow f(I)$ is an ideal in B?

$$\mathbb{Z} \xrightarrow{I} \mathbb{Q}$$
$$n\mathbb{Z} \longmapsto n\mathbb{Z}.$$

$J \subseteq B$ is an ideal, then

$f^{-1}(J) \subseteq A$ is an ideal. (verify)

<u>Def^n</u>: ① An $x \in A$ is a zero-divisor if $\exists\, 0 \neq y \in A$ s.t. $xy = 0$.

② An $x \in A$ is a nilpotent if $\exists\, n \geq 1$ s.t. $x^n = 0$.

③ An $x \in A$ is a unit if $\exists\, y \in A$ s.t. $xy = 1$.

Examples: ① In $\mathbb{Z}_6$ $\bar{2}, \bar{3}$ are zero divisors.

② In $\mathbb{Z}_4$, $\bar{2}$ is nilpotent.

③ In any non-zero element of a field is unit.

Note: A nilpotent is a zero divisor.

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